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Part - II - Boolean Algebra

2.8 Introduction

Boolean algebra is a mathematical discipline that is used for designing digital circuits in a digital computer. It describes the relation between inputs and outputs of a digital circuit. The name Boolean algebra has been given in honor of an English mathematician George Boole who proposed the basic principles of this algebra.

George Boole (1815-1864) was born to a low class family and only received an elementary school education. Despite that, he taught himself highly advanced mathematics and different languages as a teenager without any assistance. He started teaching at age sixteen, and started his own school at age nineteen. By his mid-twenties, he had mastered most of the important mathematical principles in his day.

2.8.1 Binary valued quantities:

Every day we have to make logical decisions:

1. Should I carry Computer Science book every day? Yes / No

2. 8-10 = 10 is this answer correct? Yes / No

3. Chennai is capital of India? Yes / No

4. What did I say yesterday?

The first three questions thrown above, the answer may be True (Yes) or False (No). But the fourth one, we cannot be answer as True or False. Thus, sentences which can be determined to be True or False are called "Logical Statement" or "Truth Functions". The results True or False are called "Truth Values". The truth values depicted by logical constant 1 and 0; 1 means True and 0 means False. The variable which can store these truth values are called "Logical variable" or "Binary valued variables" or "Boolean Variables" as these can store one of the two values of True or False.

2.8.2 Logical Operations:

Boolean algebra makes use of variables and operations (functions). The basic logical operations are AND, OR and NOT, which are symbolically represented by dot (.), plus (+), and by over bar / single apostrophe respectively. These symbols are also called as "Logical Operators".

2.8.3 Truth Table:

A truth table represents all the possible values of logical variable or statements along with all the possible results of given combination of truth values.